

Crime-time: how ambient light affects crime

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Abstract

This paper studies the effect of sunlight on crime, taking advantage of a daylight saving time (DST) policy. We find a 30% decrease in robberies when the DST transition increases the amount of sunlight by 1 h during the 7–9 p.m. period. We document a symmetric reaction when DST decreases sunlight exposure. We complement these findings by showing that the response induced by DST is not associated with a plausible demand-side response such as the population's commuting patterns and we find no substantial short-term displacement of crime. Our results show that ambient light is a key driver of criminal activity in urban spaces.

Keywords: Crime, robbery, ambient light, daylight saving time

JEL classifications: K42, R41, D01

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1. Introduction

Popular wisdom often identifies darkness with criminal activity. For example, in Charles Dickens' novels, crime usually happens under the so-called 'cover of darkness'. This association is not confined to fiction. As a young lawyer, Abraham Lincoln tried a famous case in which the alleged presence of light was crucial. The future President of the USA discredited one of the key testimonies against his defendant, showing that the light provided by the moon would not have allowed the witness to identify the murderer from around 100 yards away, as he had previously declared.¹ Despite this belief, empirical evaluations documenting this effect are scarce.²

This study presents several estimates of daylight's effect in a populated, dense, urban area, specifically Santiago, Chile. We find that reducing (increasing) the amount of ambient light in the evening substantially increases (reduces) criminal activity. We take advantage of the local daylight savings time (DST) policy, which imposes exogenous variation in ambient light during certain hours of the day. In particular, we find that in the vicinity

1 By using an almanac, Mr. Lincoln showed that the moon had set before the time of the incident. For details, see Henry Fonda and John Ford's film *Young Mr. Lincoln* (1939).

2 Surprisingly, Chalfin et al. (2019) is the only available study of the effect of ambient light on crime that uses experimental variation in the amount of ambient light. They find that communities in New York City that were assigned more public lighting experienced sizable crime reductions.

of the DST transition, robbery incidents decrease by 30% when DST increases the amount of sunlight by 1 h during the 7–9 p.m. period.

Similarly, we find a 30% increase in robberies when the DST transition sharply decreases daylight during the same time of day. Our results are remarkably robust to several validity checks, including two different sources of exogenous variation: a regression discontinuity design that exploits the 1-h change in ambient light exposure in the weeks around the DST transition and a difference-in-differences strategy that compares crimes reported during the same calendar period for years when the DST transition happened at different times.

We use a rich administrative database provided by the Chilean government and collected by the national police (Carabineros de Chile). This database's main advantages are that it covers the full universe of reported crimes and that it is collected by a centralized police agency. Both factors are crucial for comparability purposes (e.g. comparing different years and times of day within a large urban area). We have detailed information about each crime incident (day, time and crime type) for the 2005–2010 period in Santiago, which is the main Chilean metropolitan area.

This paper makes several contributions to the literature. We provide new estimates of the effect of ambient light on criminal activity.³ We build heavily on [Doleac and Sanders \(2015\)](#), who offer the most rigorous study of the effect of variation in ambient light caused by DST on crime, but in a notably different setting.⁴ Unlike [Doleac and Sanders \(2015\)](#) which examines primarily rural areas, which potentially limits the external validity of their results, our data cover a large and densely populated urban area, better reflecting both where people live and where crime happens. The National Incident-Based Reporting System (NIBRS) is a rich database of 558 jurisdictions in the USA, which covers a total population of approximately 22 million for the years 2005–2008. However, most jurisdictions in the NIBRS survey are in rural and low-density areas. To the extent that urban and rural areas differ in terms of criminal activity,⁵ and especially in the way ambient light shapes criminal activity, the [Doleac and Sanders \(2015\)](#) estimates do not necessarily reflect the true effect of light in high-density urban areas where most people live.⁶ This is important given that there are several mechanisms at play that could drive the empirical results. For example, if public lighting better serves urban rather than rural areas, the effect of ambient light on crime can be attenuated in urban areas. Consequently, examining the results in a highly urbanized city such as Santiago can be more informative for many large

3 [Van Koppen and Jansen \(1999\)](#) offers one of the first attempts to measure the effect of ambient light on criminal activity by comparing commercial robberies in The Netherlands across seasons of the year when sunlight exposure varies considerably. However, because they rely on OLS analysis using observational data from both summer and winter, their estimates are subject to a set of potential confounders. Another closely related study is [Chalfin et al. \(2019\)](#) who conduct the first experimental evaluation of the effect of street lighting on crime. Although their intervention is very different than the 1-h variation in ambient light imposed by DST, they also find a strong negative relationship between light and criminal activity; in particular, they find that provision of street lighting led to a at least 36% reduction in outdoor crime during night hours.

4 A pioneer study that exploits sharp variations in visibility imposed by DST policy is [Grogger and Ridgeway \(2006\)](#) in the context of police stops (see [Horrace and Rohlin, 2016](#); [Pierson et al., 2020](#), for more recent implementations).

5 See, for example, [Glaeser and Sacerdote \(1999\)](#), [Grogger and Willis \(2000\)](#) and [O'Flaherty and Sethi \(2015\)](#).

6 Due to urban agglomeration, studies that focus on a single metropolitan area can analyze the effect of more granular events. For example, [Montolio and Planells-Struse \(2016\)](#) examines the impact of football matches on the temporal evolution of crime in Barcelona.

metropolitan areas across the world.⁷ With a population of over 6 million people, Santiago is the sixth largest city in Latin America and is the 28th most densely populated city on earth.⁸

Our findings confirm some of the results in [Doleac and Sanders \(2015\)](#). However, unlike their work, we find a consistent response when DST imposes a reduction in ambient light.⁹ The fall DST transition offers a natural way to test the relationship between ambient light and crime. In a previous and extended version of their paper, [Doleac and Sanders \(2012\)](#) also find a small but barely significant reduction in robbery during the evening. It is hard to reconcile this finding through the sunlight mechanism; however, since the fall DST transition imposes a reduction in ambient light at this time.¹⁰ [Doleac and Sanders \(2015\)](#) claim that the coincidence of the fall DST transition with Halloween is confounding the relationship between criminal activity and sunlight due to unusual Halloween activity that may prevent them from detecting a consistent effect. In our case, we do detect a significant increase in robbery when DST imposes a reduction in ambient light in the evening.

Importantly, we also find consistent symmetric responses in robbery at sunrise hours for each DST transition. Finally, in terms of magnitude, the variation in the response is symmetric, which reinforces our finding of the negative relationship between ambient light and criminal activity.

We also analyze potential temporal re-allocations of criminal activity associated with the variation in sunlight imposed by the DST policy. We find no evidence of substantial crime displacement to other times of the day. Although we cannot rule out all other possible types of displacement (e.g. types of crimes, level of violence and victim type), our results suggest that short-term displacement within the day is at most weak and not consistent when compared with the opposite DST transitions. Only a portion of the sunset crime reduction, if any, seems to be compensated for with an increase in night crime; this coincides with the fact that the overall daily variation in light associated with DST is similar or larger in absolute value to the variation observed during sunset hours. In a way, the lack of evidence supporting temporal displacement as well as the symmetric reaction to ambient light in different periods of the year is consistent with a broad strand of the literature that emphasizes the opportunistic nature of criminal activity ([Clarke, 1995](#)).

By focusing on a single large city, we are able to discuss an additional issue regarding the effect of ambient light on crime. We analyze the degree to which our reduced-form results on criminal activity could be driven by other types of responses. In particular, we examine an aspect of victims' behavior that, according to [Cook et al. \(2011\)](#), has been 'largely neglected in the economics literature' ([Cook et al., 2011, 10](#)). Using high-frequency data on Santiago's subway ridership, we analyze whether the variation in ambient light is also associated with changes in ridership. We do not find significant variation induced by the DST transition. Of course, this piece of information cannot

7 Both areas differ in many regards, but a significant one is the pattern of criminal activity. The average annual murder rate (incidents per 100,000 inhabitants) is considerably lower in Santiago (2.05) than in [Doleac and Sanders \(2015\)](#)'s database (5.14). Meanwhile robbery rates are reversed: 508 per 100,000 in Santiago versus 120 for [Doleac and Sanders \(2015\)](#).

8 Source: City Mayors' ranking at www.citymajors.com. Extracted on 10 March 2021.

9 We distinguish several time periods within a single day: sunrise (6–8 a.m.), daytime (8 a.m.–7 p.m.), sunset (7–9 p.m.) and nighttime (9 p.m.–6 a.m.).

10 However, they do report a positive and economically significant reduction in shoplifting and burglary. See Table 7 from [Doleac and Sanders \(2012\)](#) for details on each crime category.

rule out other possible endogenous responses associated with victim or police behavior; however, this information is at least suggestive of the fact that potential offender or supply-side responses, rather than victims' endogenous reactions, could be driving our results.

2. Empirical analysis

2.1. Data and the DST transition

Our analysis is focused on Santiago, Chile, a city of more than 6 million inhabitants. We use administrative data of all crime reported to police between 2005 and 2010. Each crime report contains information about the time and location where the crime was perpetrated, as reported by the victim and crimes are classified into 10 different categories. Crime reports are collected by the Chilean police, which is a very centralized organization (Carabineros de Chile). [Table 1](#) summarizes the data we use by type of crime and period of the year.

We also collect information on the actual DST implementation for each year. This is from the 1489 Act records (Decreto 1489). In 'normal' years, DST transitions occur after the second Saturday of March (fall transition) and October (spring transition). In addition, we collected data on the exact sunset and sunrise times in 2007. This lets us precisely measure the sharp variations in terms of sunlight exposure associated with a particular DST transition during a 'normal' year. [Figure 1](#) shows the daily evolution of sunset (blue line) and sunrise hours (red line) in 2007. [Figure 1](#) motivates our identification strategy that relies on the sharp change in sunlight exposure at different times of the day that happen, in opposite directions, at two separate times of the year.

Importantly, we have two non-normal years (2008 and 2010) when the March transition was delayed for 3 weeks due to natural causes: the 2008 drought and the 8.8 Richter scale earthquake (27 February 2010) that primarily affected central Chile. We will come back to these two exceptions when we describe our identification strategies.

The empirical challenge of this paper is to analyze the relationship between daylight and criminal activity. As we discussed earlier, it is hard to disentangle the extent to which sunlight modifies everyday behavior, which may affect opportunities for crime and the behavior of potential offenders. Rather than make an attempt at a complete description of criminal behavior where sunlight exposure may be an important factor, we rely on exogenous variations in sunlight caused by DST policy that offer an opportunity to empirically estimate the causal effect. We estimate the effect of the DST transition on crime throughout the day for the two periods of the year where DST changes. In addition, we offer a complementary identification strategy that also relies on exogenous variation in ambient light across the calendar year.

First, we describe a regression discontinuity design that takes advantage of the sharp variation in daylight exposure that DST produces around sunrise and sunset hours. We use this approach to test whether an increase/reduction in daylight exposure causes a reduction or an increase in crime. We complement this approach by implementing a second identification strategy which relies on a different source of variation due to an exogenous delay in the DST transition. As mentioned before, in 2008 and 2010, the government decided to delay the fall transition (March) by three weeks. We believe that this exogenous variation offers another research opportunity to analyze the effect of daylight on crime.

Table 1. Daily average number of incidents by crime category

Crime	Fall	Spring	Rest of the year
Robbery	86.06	94.37	88.70
Larceny	30.03	33.92	30.61
Vehicle theft	34.10	38.91	35.80
Theft from MV	65.89	72.94	68.95
Burglary with people	52.92	58.53	59.83
Burglary without people	31.04	35.60	34.11
Other robberies with the use of force	5.01	6.06	5.33
Theft	87.45	98.96	91.48
Murder	0.34	0.42	0.36
Rape	2.73	2.77	2.63

Notes: The table shows daily average number of incidents for each period of the year. We divided all incidents into three exclusive categories: Fall (March), Spring (October) and rest of the year which averages across all months except March and October. We follow Chilean police crime classification. Most crime definitions are self-explanatory. Theft is defined as the appropriation of someone else's property. Theft takes place with no violence, intimidation or force. Larceny refers to cases when the perpetrator operates by surprise, pretending to fight or performing maneuvers to confuse potential targets. 'Other robberies with the use of force' requires the use of force, but it excludes crimes with direct involvement of a human victim.

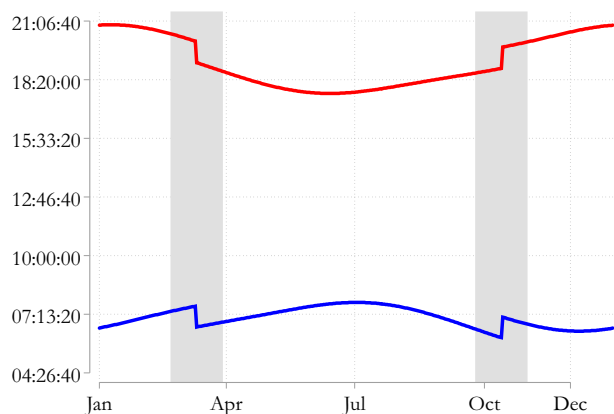


Figure 1. Sunrise and sunset hours in 2007. Red and blue lines, respectively, represent the exact sunset and sunrise times for each day of the year. Shaded areas represent each of the periods of analysis considered in our identification strategy (see Section 2.2): fall in mid-March and spring in mid-October.

Source: Astronomical information taken on 2 February 2016 from www.tutiempo.net.

2.2. Sharp regression discontinuity design

The potential outcomes framework or the so-called Neyman–Rubin causal model offers a simple way to specify why, under certain circumstances, a regression discontinuity design yields a causal estimate of a particular effect (Imbens and Lemieux, 2008). Its basic identification assumption is that the conditional expectation functions of potential outcomes are continuous in the vicinity of a certain cutoff. Our dependent variable Y_i represents the amount of criminal activity observed during a particular period of the day and X_i is a temporal measure that indicates proximity to the DST transition. Formally, we can write

$$E[Y_i(0)|X_i = x] \text{ and } E[Y_i(1)|X_i = x] \text{ are continuous in } x. \quad (1)$$

Thus, we can estimate the average treatment effect ρ around a certain point c as follows:

$$\rho = \lim_{x \rightarrow c^+} E[Y_i|X_i = x] - \lim_{x \rightarrow c^-} E[Y_i|X_i = x] = E[Y_i(1) - Y_i(0)|X_i = c]. \quad (2)$$

We implement this strategy since we know the exact rule defining the treatment assignment, which in this case is determined by the time schedule imposed by the DST policy. The continuity assumption requires smoothness in a small neighborhood of the DST transition, so any discontinuity in the conditional distribution of the outcome at the threshold value can be interpreted as evidence of a causal effect (Imbens and Lemieux, 2008). In this case, the treatment can be defined as a sharp variation in daylight exposure, which is precisely determined by the DST transition. In particular, we focus on the 1-h variation imposed by the biannual DST transition. In terms of sunlight exposure, for each DST transition there are two times of day that are exposed to this source of variation; we call them sunset (19:00–20:59) and sunrise (6:00–7:59) hours. There is no reason to believe that other times of day (nighttime and daytime) are exposed to this particular treatment during the same period. Following Angrist and Pischke (2008), we run several regressions depending on the period of the day (h) as follows:

$$\log(\text{Crime}_{ih}) = \alpha_h + \beta_{1h}X_{ih} + \beta_{2h}\text{Post}_{ih}X_{ih} + \rho_h\text{Post}_{ih} + \omega_{hd(i)} + \psi_{ht(i)} + \epsilon_{ih}. \quad (3)$$

The dependent variable $\log(\text{Crime}_{ih})$ measures the log of total crimes for a particular day i during the daytime period h . Unless otherwise specified, Crime_{ih} includes all robberies, larcenies, thefts, vehicle thefts, burglaries, murders and rape incidents. The running variable X_{ih} indicates the number of days before and after the DST transition and is centered to zero, meaning the actual day of the DST transition for each year t in the data. Post_{ih} is an indicator function of whether day i was exposed to the DST transition or not and we also include an interaction term $\text{Post}_{ih} \times X_{ih}$ to control for any change in slope on either side of the threshold. In some specifications, we include three cloud indicators based on a unit of measurement used to describe the amount of cloud cover (oktas). We classified each day based on three categories: clear sky (1 or 2 oktas), sky half cloudy (3, 4 or 5 oktas) and cloudy (6, 7 or 8 oktas).¹¹ In order to have a more flexible function, we consider additional functional forms such as a quadratic specification of the running variable, X_{ih}^2 , and an interacted term, $\text{Post}_{ih} \times X_{ih}^2$. We further include $\omega_{hd(i)}$ (day of the week) and $\psi_{ht(i)}$ (year fixed effects) to better approximate the cyclical structure of crimes and finally ϵ_{ih} represents the idiosyncratic error term. The parameter ρ_h is our coefficient of interest which captures the effect of sunlight variation on the percentage of total crimes during a particular period of the day.

2.3. Difference-in-differences approach

As a complementary analysis, we estimate the effect of this sharp variation in daylight exposure using a second identification strategy. Here, we exploit another source of exogenous variation, namely a delay in the DST transition. In particular, we take advantage of

11 Data were extracted from the Chilean meteorological agency on 10 December 2021 [link].

the fact that in two specific years the Chilean government decided to delay the DST fall transition, which by law is supposed to occur after the second Saturday of March. Under the assumption that this decision was orthogonal to our dependent variable, we can build a natural counterfactual for our treatment group determined by those years where the DST fall transition happened as defined by law.

A reasonable concern about the RD identification strategy is that crime incidence may also be affected by the time of year. This can be particularly relevant for the estimates based on the fall season (DST transition in March). In Chile, March is the first ‘regular’ month of the year and the first 2 weeks see significant changes in social and economic life. The academic school year and many jobs start in these 2 weeks, right after the end of a vacation period during January and February. If people are adjusting to new schedules during early March, the difference between these 2 weeks and the following two may capture more than simply the effect of the DST schedule. In that sense, the RD estimates might be biased.

The difference-in-differences strategy offers a robustness check for this concern since it provides additional variation in the timing, allowing us to examine the effect at other times of the year. Specifically, in 2008, due to a bad drought, the government decided to delay the DST transition by 3 weeks with the hope of reducing national energy consumption. Similarly, in 2010, after a bad earthquake that mainly affected cities south of Santiago, the government likewise decided to postpone DST by 3 weeks in order to help families, volunteers and organizations dealing with the earthquake, providing them increased sunlight in the evenings.

We follow [Doleac and Sanders \(2015\)](#), taking advantage of the variation in the week of the year when DST was implemented and the variation in the impact of DST across different hours of the day. For this case, we restrict the sample to the earliest DST transition in March in the data; between March 8 (in 2005) and March 29 (in 2008). It is important to keep in mind that this particular source of variation is available only for the fall DST transition, which occurs in March. During this particular transition, the amount of sunlight sharply decreases during sunset hours. Obviously this comes with a corresponding increase during sunrise hours. We collapse all the data to day-by-sunset level, where ‘sunset’ represents a 2-h period when the sunset time actually happened in that year. Our basic regression can be described as follows:

$$\log(\text{Crime}_{it}) = \alpha + \beta_1 \text{DST}_{it} + \beta_2 \text{Sunset}_{it} + \gamma \text{DST}_{it} \times \text{Sunset}_{it} + \epsilon_{it}, \quad (4)$$

where the dependent variable represents the log of total crimes in a particular day-hour period and DST_{it} indicates whether observation i is in years with a regular DST transition (2005, 2006, 2007 and 2009), as opposed to the ‘control years’ of 2008 and 2010. In other words, β_1 accounts for average permanent differences between years with and without the default DST policy. Sunset_{it} indicates whether observation i happens during ‘sunset-hours’, as opposed to any other time, so it enters this model as a common-trend effect for treatment and control groups as is usual in difference-in-differences specifications. γ represents our parameter of interest; it captures the double difference between treatment and control groups and sunset-hours versus other times of the day.

3. Results

Our main results are presented in three parts. First, we present a set of stylized facts comparing the distribution of criminal activity across hours of the day for a short period of

time before and after the DST transition. These results emphasize the change in average daily crimes a few days before and after the transition that occurred during hours of the day effectively exposed to the sunlight variation. Then, we move to RD estimations beginning with a graphical representation for our preferred RD estimate. We present regression coefficients under different model specifications. Finally, we give the difference-in-differences estimates. Results for other times of day and periods of the year are fully reported in the extension section and in the [Online Appendix](#).

3.1. Graphical analysis

The first feature of crime distribution over time that we want to highlight is its variation during an average day. [Figure 2](#) plots a histogram of crime over hours of the day for two different periods: the 3 weeks before (blue bars) and after (empty bars) the spring DST transition. For the two periods, the distribution of crime incidents is very similar and shows a general pattern with a small proportion of incidents during nighttime hours (midnight to 6 a.m.), followed by a period of an increasing rates until 1 p.m. After 1 p.m., the rate of incidents remains stable until 6 p.m. The period between 7 p.m. and 9 p.m. displays the highest crime rate. If the variation in light is affecting criminal activity, we may expect to see an important variation during those specific hours. Indeed, we observe a sharp reduction in crime incidents around 7 p.m. and 8 p.m. with increased light, associated with DST. [Figure A1 in the Online Appendix](#) shows the same pattern for the fall (March) DST transition, but in the opposite direction. We will later discuss the implications of this symmetric pattern across spring and fall DST transitions. This pattern is also remarkably consistent across different specifications.¹²

Our primary focus is the discontinuity at the DST transition. Based on the basic comparison described in [Figure 2](#), we expect the largest variation in the period between 7:00 p.m. and 9:00 p.m. Similarly, we may expect variations around the sunrise hours. In addition, analyzing similar responses for other periods of the day that are not affected by variations in ambient light are also relevant since they may offer a natural measure of any *placebo* effect. Based on the variation in ambient light imposed by the DST policy, we define four relevant periods of analysis: nighttime (9:00 p.m. to 5:59 p.m.), daytime (10 a.m. to 5:59 p.m.), sunset (7:00 p.m. to 8:59 p.m.) and sunrise (6:00 a.m. to 7:59 a.m.). We later relax this period definition by examining the responses across each hour of the day.

[Figure 3](#) plots the residuals from a regression that adjusts for the variation in robbery incidents relative to the date of DST transition for each of the relevant periods of analysis, controlling for year and day-of-week fixed effects.¹³ We observe a substantial change at the DST change in [Figure 3\(c\)](#), which restricts the comparison to sunset hours.

12 In the [Online Appendix](#), [Figures A2](#) and [A3](#) show similar histograms for each year and its DST transitions. Finally, the [Online Appendix](#) further includes [Figures A4](#) and [A5](#), again histograms of crime by time of day but modifying the window period.

13 In the [Online Appendix](#), we include several additional specifications that may alleviate concerns about the specification choices made in [Figure 3](#). First, [Online Appendix Figures A6](#) and [A7](#) show the discontinuous change in daily robberies for each of the relevant periods of analysis by comparing the predicted values estimated from a regression that includes year and day-of-the-week fixed effects. In each case, we observe a clear cyclical pattern of crime activity with a dramatic change right after the DST transition only during the sunset period. We also include figures with a quadratic and a flexible polynomial fit on either side of the threshold. For the spring DST transition, see [Online Appendix Figures A9](#) and [A10](#), and for the fall see [Online Appendix Figures A12](#) and [A13](#).

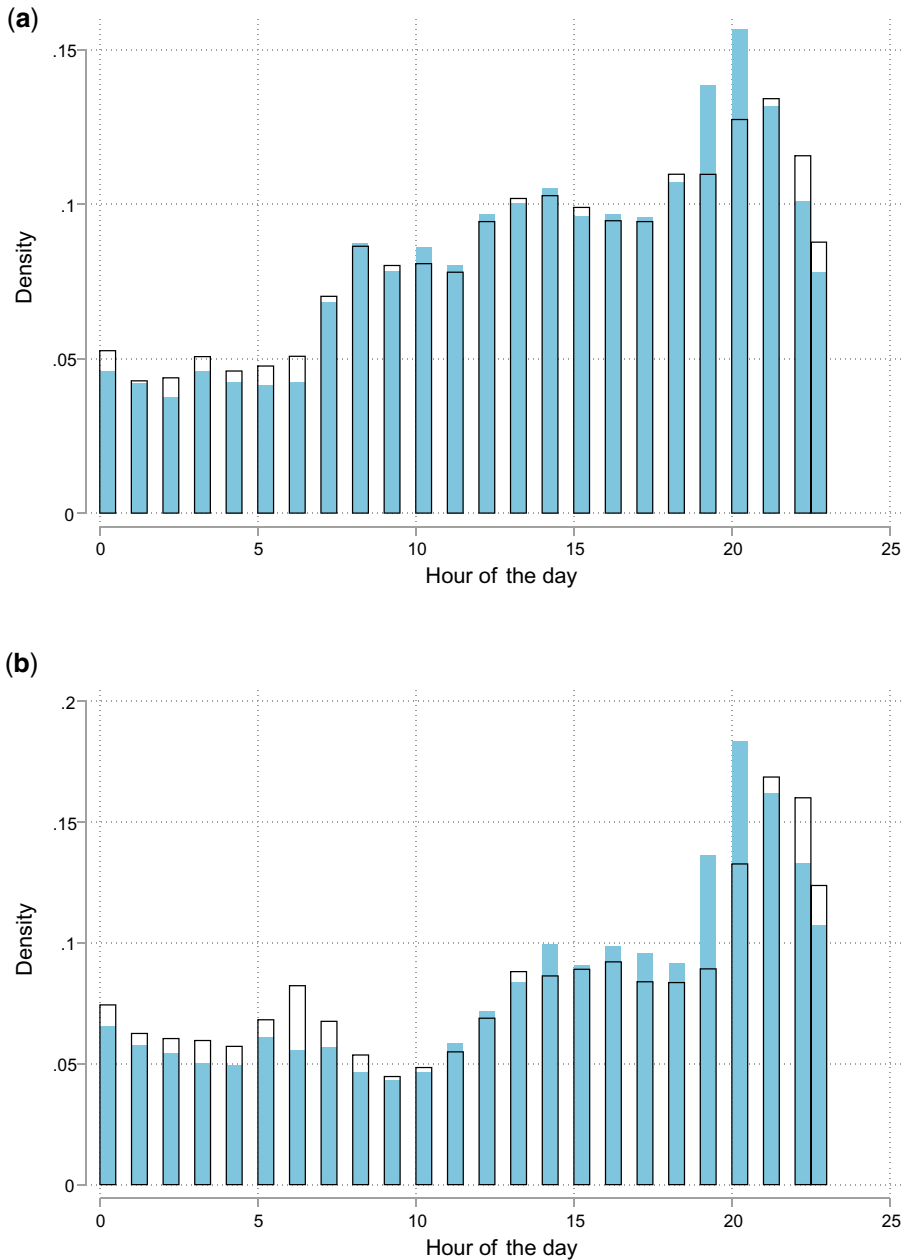


Figure 2. Distribution of crime by hour of day around spring DST transition. (a) All crimes. (b) Robbery. Histogram of crime reports by hour of day and periods of analysis before and after the spring DST transition. Blue-shadowed areas represent incidents reported 3 weeks *before* the DST transition, while the empty bars show the average over the 3 weeks *after* the DST transition. The spring DST transition usually takes place after the second Saturday in October.

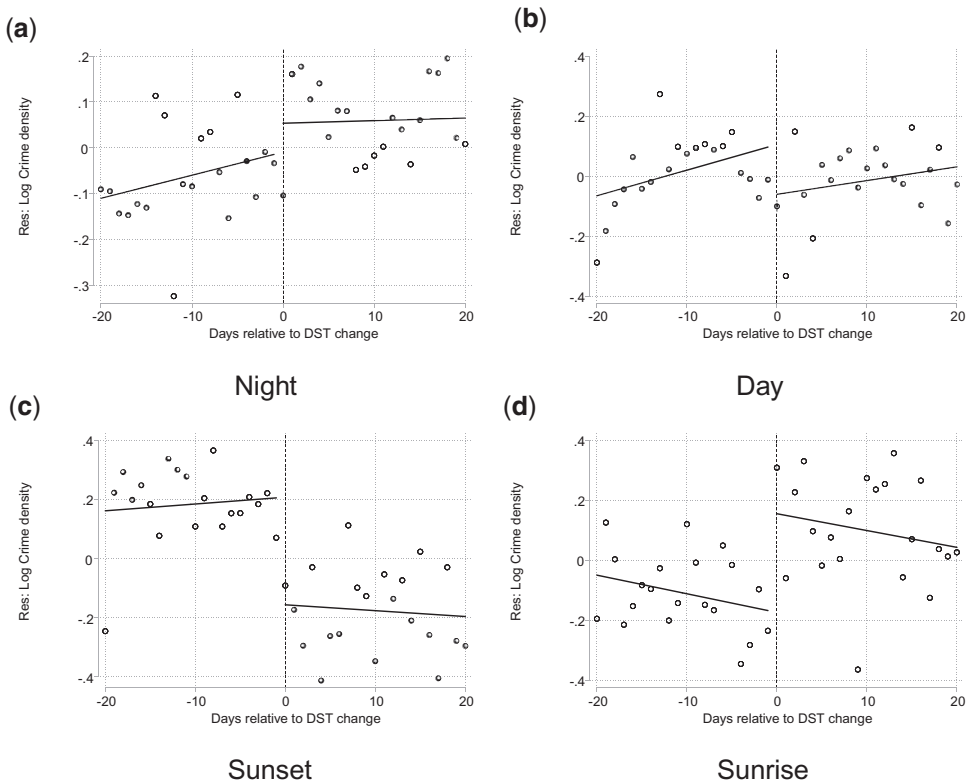


Figure 3. Robbery incidents during spring DST transition by daytime period. (a) Night. (b) Day. (c) Sunset. (d) Sunrise. Each figure corresponds to a scatterplot of residuals from separate regressions using average log-robbery incidents, controlling for year and day-of-week fixed effects for each period of time and on both sides of the threshold. Lines represent linear fits at each side of the threshold. The horizontal axis is the number of days away from the DST transition and the sample is restricted to 21 days on both sides of the threshold.

We interpret the sharp discontinuity at the threshold in [Figure 3\(c\)](#) as the effect of the change in ambient light on criminal activity. Specifically, we observe more than a 30–40% reduction in robbery associated with increasing by 1 h the amount of ambient light exposure during sunset hours (7 p.m. to 9 p.m.). To clarify by example, on Saturday 10 August 2005 sunset was at 6:49 p.m., while on the following day, Sunday 11 August 2005, sunset was at 7:49 p.m. Conversely, during sunrise hours, we experience a sharp decrease of 1 h of sunlight which corresponds to a 30–40% increase in robbery. Importantly, we do not observe similar discontinuities in criminal activity for other periods of the day where no variation in ambient light occurs.

In the [Online Appendix](#), we include similar results exploiting the variation in ambient light induced by the fall DST transition. Interestingly, we find similar results in magnitude but with the opposite sign. We interpret this finding as consistent with the fact that variation in ambient light is directly affecting the amount of criminal activity observed. As in [Figure 3](#), [Figure A8](#) in the [Online Appendix](#) plots the linear fit of the residuals on either side of the threshold. In addition, [Online Appendix Figures A7–A13](#) display in different ways the discontinuity around a threshold defined by the DST transition for the four periods of the day.

3.2. Basic estimates

3.2.1. RD estimates

Table 2 shows the results for our variable of interest, which represents the days after the DST transition. In panel A, we show the coefficient for the spring regression and similar results are displayed in panel B for the fall. In both panels, under different functional specifications, a 30% change in criminal activity results from the change of an hour of sunlight. In this case, the effect of ambient light is remarkably symmetrical. In panel A, where *post* refers to an increase in the provision of ambient light, we observe a 30% decrease in robbery, whereas in panel B we observe a mirrored effect; a decrease in ambient light increases crime by a similar amount. Additional results for the other relevant periods of the day are shown in Online Appendix Table A3. Our preferred estimation considers a sample that exclude days related to the September holiday season (Chilean national day). Interestingly, we do not see significant variation that can be attributed to DST during other periods of the day for either transition.¹⁴

3.2.2. Difference-in-differences estimates: fall DST transition

We now evaluate the robustness of the results under a different identification strategy. As we previously discussed, the fact that the DST transition takes place at a particular period of the calendar year can bias our RD coefficients—there may be other changes happening at the same time of year that are also affecting criminal activity. Although limited, this concern could be important during the fall DST transition that usually takes place after the second Saturday in March, which is a period of the year where people are typically learning to adapt (or readapt) to new time schedules. Our difference-in-differences estimates address that concern by capturing the double difference between years where the DST transition was implemented as usual and years when there was a delay of 3 weeks.

We estimate specification (4), after collapsing the data to the day-by-sunset level, where ‘sunset’ represents a 2-h period when sunsets actually happen. We additionally repeat this procedure, considering ‘sunset-hours’ or the relevant period each hour of the day, which directly offers a falsification test for the effect of daylight variation on crime. Figure 4 summarizes all these estimates with their respective 95% confidence intervals.

As we can see in Figure 4, it is very clear that there is no significant effect except for the estimates that are based on sunset hours. These estimates represent a 25% increase in robbery (and a 15% increase in total crime), which is slightly smaller in absolute value than our previous RD specification based on the spring DST transition (30% and 20% decrease, respectively).

3.2.3. Estimates by crime type

Table 3 presents separate estimates for each crime category and period of the day. The criminal categories are at the most disaggregated level available in the data. There are many possible ways ambient light can differently affect different crimes. Among the most crucial distinctions for the purposes of our identification strategy is the accuracy of the reported time for each type of crime. Since we are using reports that are mostly made by victims, the accuracy of the reported time is influenced by the type of offense they

14 Similar results for all crimes aggregated can be found in Online Appendix Tables A1 and A2.

Table 2. RD estimates at sunset hours by DST transition

Dependent variable: log (robbery)	(1)	(2)	(3)	(4)	(5)
Panel A: Spring					
Post	−0.366*** (0.1033)	−0.325*** (0.0938)	−0.353*** (0.0940)	−0.334* (0.1302)	−0.408** (0.1343)
Observations	248	248	243	248	243
R ²	0.20	0.42	0.44	0.42	0.44
Panel B: Fall					
Post	0.219* (0.0979)	0.296** (0.0974)	0.293** (0.0982)	0.378** (0.1413)	0.367** (0.1397)
Observations	220	220	220	220	220
R ²	0.17	0.34	0.36	0.36	0.37
DoW FE	N	Y	Y	Y	Y
Year FE	N	Y	Y	Y	Y
Cloud indicators	N	N	Y	N	Y
Quadratic	N	N	N	Y	Y

Notes: Coefficients from separate regressions using Equation (3) at sunset hours. Running variable is days before and after spring DST transition. Post indicates days after the DST transition. DoW (day of the week) and Year indicates whether these fixed effects were included in the regression. Cloud indicators are a set of three indicator variables reflecting different levels of cloudiness. Both panel coefficients are estimated considering 3 weeks before and after the DST transition. In panel A, we exclude days associated with Chilean national day. In panel B, we exclude February days since they correspond to summer vacations in Chile. Robust standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$.

suffered. For instance, a victim can clearly recall the time when he/she was robbed, while in the case of a ‘burglary without people’ there may need to be an estimation based on some basic facts (last time someone was present, when they noticed signs of the burglary, etc.). In addition, we could expect that crimes that usually take place outdoors would be more responsive to changes in ambient light. For example, while most robberies, larcenies and vehicle thefts take place outdoors, only a small portion of thefts do so.

Table 3 shows coefficients for both DST transitions. In panel A, we display results for spring and results for fall are in panel B. At first glance, almost all of the estimates are not significant except for the case of robbery. We observe that robbery is likely driving our previous estimates when we pool all crime together, considering both its high share of overall crime and its sensitivity to time. In fact, relative to all crimes, we find larger responses for robbery within each time period affected by a change in ambient light.

4. Extensions

In this section, we discuss two additional features of our findings.¹⁵ First, we carefully analyze our results to discuss the degree to which we can detect temporal reallocation of

15 In Online Appendix sections A2.1 and A2.2, we show that our results are robust to several validity checks such as considering placebo DST transitions across the calendar year, modifying the unit of analysis (daily or weekly) and the RD bandwidth definition.

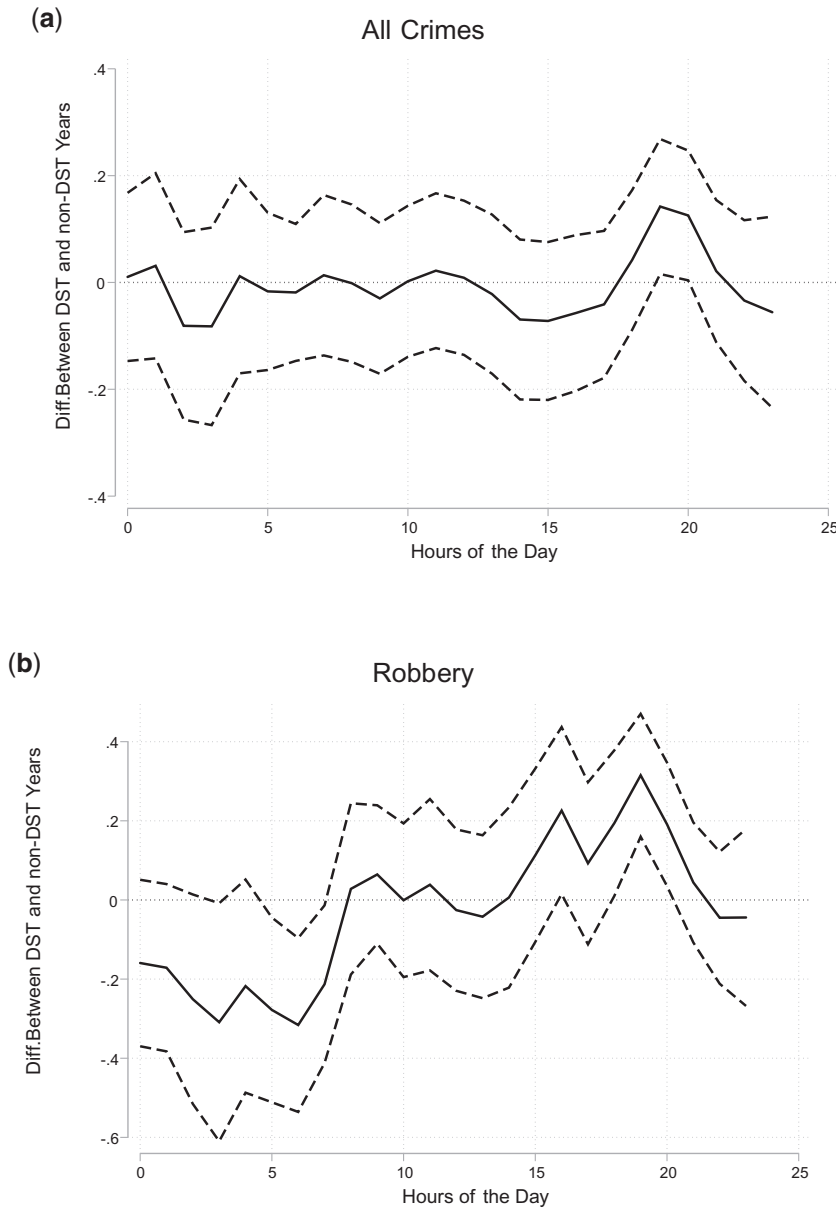


Figure 4. Difference-in-differences estimates by hour of day: fall DST transition. (a) All crimes. (b) Robbery. The figure represents hourly estimates of 24 difference-in-differences regressions as defined by Equation (4). Each regression is estimated using a 2-h window period and the coefficient is plotted at the initial hour of the period. The sample is restricted to the period between the earliest March DST transitions (between 8 March and 29 March). Panel (a) displays results for all crimes and panel (b) shows results restricting the sample to robbery incidents.

Table 3. RD estimates at each time of the day by crime type and period of the year

	Panel A: Spring				Panel B: Fall			
	(1) Night	(2) Day	(3) Sunset	(4) Sunrise	(5) Night	(6) Day	(7) Sunset	(8) Sunrise
All crimes	0.0460 (0.0392)	−0.0890 (0.0456)	−0.207*** (0.0584)	0.0385 (0.0662)	−0.00819 (0.0521)	0.0777 (0.0570)	0.170* (0.0714)	−0.0628 (0.0766)
Robbery	0.0555 (0.0513)	−0.153* (0.0645)	−0.325*** (0.0938)	0.280* (0.115)	0.0880 (0.0656)	0.0794 (0.0635)	0.296** (0.0974)	−0.247 (0.135)
Larceny	−0.00180 (0.120)	−0.222** (0.0848)	−0.0673 (0.130)	0.0383 (0.153)	0.0106 (0.150)	0.0478 (0.103)	0.309 (0.159)	−0.263 (0.159)
Vehicle theft	0.100 (0.0814)	−0.170* (0.0784)	−0.221 (0.130)	−0.0234 (0.158)	−0.192 (0.102)	0.0215 (0.0956)	0.129 (0.150)	0.0468 (0.167)
Burg w/people	0.110 (0.0688)	−0.0243 (0.0679)	−0.247 (0.141)	−0.129 (0.156)	0.0665 (0.0729)	0.0696 (0.0711)	0.166 (0.115)	0.115 (0.162)
Burg w/o people	−0.0174 (0.0691)	0.0208 (0.0571)	−0.272* (0.130)	−0.135 (0.161)	−0.162 (0.1000)	−0.00350 (0.0736)	0.225 (0.163)	−0.0832 (0.142)
Other robbery	0.0102 (0.0974)	0.0512 (0.0841)	−0.258 (0.162)	−0.0398 (0.146)	−0.0576 (0.108)	0.0390 (0.0820)	0.0308 (0.171)	0.236 (0.142)
Theft	−0.161 (0.174)	−0.227 (0.159)	−0.0914 (0.164)	−0.249 (0.162)	0.00434 (0.173)	0.00959 (0.175)	−0.184 (0.157)	0.0489 (0.125)

Notes: The table shows coefficients from separate regressions following Equation (3) by type of crime and period of the day. Dependent variable is log (crimes). Running variable is days before and after spring DST transition. Post indicates days after the DST transition. Both panel coefficients are estimated considering 3 weeks before and after the DST transition. In panel A, we exclude days associated with Chilean national day. In panel B, we exclude February days since it corresponds to summer vacations in Chile. Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$.

criminal activity (short-term displacement). Second, we analyze the extent to which our findings are driven by a possible demand-side response regarding potential victims’ behavior. Particularly, we study how DST affects the commuting patterns using high-frequency data from Santiago’s subway network.

4.1. Evaluating possible temporal reallocation of criminal activity

In the previous section, we find no significant response in criminal activity at times of day when the DST transition did not affect the amount of light. In a way, this evidence alone suggests there is no short-term displacement in crime. But more generally, we would like to evaluate whether the effect we observed at sunset hours generally corresponds with an offsetting reaction during the rest of the day. Evidence of displacement is important since it can substantially alter the evaluation of a particular policy. An interesting case in this regard is Jacob et al. (2007), which finds substantial temporal displacement of crime by exploiting weather shocks that significantly alter the amount of criminal activity across US cities. They present a dynamic model of criminal behavior and show that for long periods (across weeks), displacement is consistent with wage fluctuations with a meaningful income effect that persists across periods.

If no temporal displacement occurs in our situation, we can expect that the effect at sunset hours—when a considerable portion of criminal activity takes place—strongly

influences the overall crime variation induced by the DST policy. Thus, RD estimates using Equation (3), with overall daily crime incidents as the dependent variable, should yield a similar variation in the number of reported crimes.

Table 4 compares our previous estimates at sunset hours and the overall aggregate response at the daily level. In panel A, we compare all crimes whereas panel B focuses on robbery, the crime with the largest response to changes in ambient light. We also include the implied change in terms of additional incidents that can be associated with the regression estimate, which is calculated by multiplying the mean of the dependent variable with the corresponding effect. Although in some specifications the effect is not significant at conventional levels, we observe that in all cases the overall daily effect and the partial effect at sunset hours have similar signs. Moreover, in all cases, the daily variation exceeds the implied change in incidents at sunset hours in absolute value. This suggests that any changes happening during the rest of the day can hardly compensate for the changes during sunset hours.

4.2. Evaluating a response in commuting patterns

Most empirical studies on the economics of crime literature take victims' behavior as given. But potential victims can affect criminal activity in different ways, especially when they perceive changes in the risk of suffering a crime. They can avoid certain areas or particular periods of the day, harden a particular target or offer a higher resistance level when attacked (Dominguez, 2020). Ideally, we would incorporate a full specification of victims' behavior for every hour of the day into our estimation. Considering that most criminal activity takes place when people return home in the evening, we analyze the extent to which variation in victims' commuting drives our estimates. To evaluate this, we use high-frequency data on ridership on the Santiago subway and run the following regression for every period of the day.

$$\log(M_{ih}) = \alpha_h + \beta_{1h}X_{ih} + \beta_{2h}\text{DST}_{ih}X_{ih} + \rho_h\text{DST}_{ih} + \omega_{hd(i)} + \psi_{ht(i)} + \epsilon_{ih}. \quad (5)$$

In this equation, M_{ih} represents the number of people who entered the Santiago subway system in period h of day i . Equation (5) also controls for year and day-of-week fixed effects. The Santiago subway system transports 2.5 million passengers per day, which represents around 18% of total trips in an average day (calculations by the authors). In 2010, the Santiago subway had 108 stations with a total network of approximately 100 km. Our data include the number of commuters who entered the subway system every hour and cover the same years as our crime data.

Results from regression (5) are displayed in Table 5. We matched the data for the exact period of analysis we use in Table 2 and Online Appendix Table A3. There is no significant variation associated with changes in ambient light. Also, the signs of the sunrise and sunset coefficients are the same for each DST transition, which indicates that a sharp variation in ambient light does not correspond with consistent ridership changes. This suggests that our results are not fully driven by a change in commuting patterns. This finding slightly contrasts with the conclusion of Wolff and Makino (2012), who evaluate the effect of DST policy on time allocation using the American Time Use Survey.¹⁶

16 They find a modest variation associated with the DST transition. When ambient light increases during evening hours, people report spending 3% more of their time in outdoor recreational activities and reducing their TV watching by 9 min.

Table 4. Implied change in incidents by period

Time block:	Spring		Fall	
	19–21 h	0–24 h	19–21 h	0–24 h
Panel A: All crimes				
Post	−0.207*** (0.06)	−0.056 (0.04)	0.170* (0.07)	0.055 (0.05)
Mean DV	58.73	443.26	51.81	405.01
Implied change	−12.13	−25.03	8.82	22.38
Panel B: Robbery				
Post	−0.313*** (0.09)	−0.067 (0.04)	0.256** (0.10)	0.084 (0.05)
Mean DV	13.29	101.25	11.35	91.46
Implied change	−4.17	−6.82	2.91	7.71

Notes: The table shows coefficients from separate regressions following Equation (3). Dependent variable is log (crimes). Mean DV is expressed as the number of incidents reported during each period. 19–21 h indicates regressions restricted to sunset hours whereas 0–24 h indicates regressions estimated at the overall daily level. Implied change is calculated by multiplying the average number of incidents with the corresponding effect of the on crime. Robust standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$.

5. Conclusion

In this paper, we present new estimates of the causal effect of ambient light on criminal activity. We extend previous research in the literature to a highly dense and populated urban area. Building on the strategy adopted by Doleac and Sanders (2015), we validate our results using two different sources of variation and confirm the magnitude and direction of many of their findings. We find that a 1-h increase (reduction) in the amount of light at sunset hours (between 7 and 8:59 p.m.) reduces (increases) robbery by 30% and all criminal activity by 20%. Our results are robust to a variety of model specifications. Particularly, we find no significant responses associated with placebo variations across hours of the day during the DST transition or from considering placebo DST transitions across days of the year. Moreover, our results are robust to the bandwidth definition as well as the level of data aggregation (daily or weekly level).

Our findings not only confirm the literature’s previous findings in a novel setting but also reinforce the negative relationship between ambient light and criminal activity. As opposed to Doleac and Sanders (2015), our estimates for the two DST transitions are similar in absolute value and we do observe a significant increase in robbery when ambient light decreases during sunset hours. We also discuss possible short-term displacement in criminal activity potentially caused by DST policy. This is an important issue since evidence of substantive time-displacement may limit the scope for action of policies oriented to reduce crime through the use of artificial light. Although we cannot fully reject every possible dimension of crime displacement (e.g. between locations or different types of criminal activity), we find no significant evidence of displacement to other times of the day.

Table 5. RD estimates of metro ridership during DST transition

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Spring	Night	Day	Sunset	Sunrise	Night	Day	Sunset	Sunrise
Post	0.00431 (0.0354)	-0.102 (0.0684)	-0.0742 (0.0549)	-0.291 (0.216)	0.0167 (0.0563)	-0.204 (0.105)	-0.133 (0.0827)	-0.889** (0.328)
<i>N</i>	248	248	248	248	248	248	248	248
<i>R</i> ²	0.951	0.874	0.894	0.848	0.952	0.881	0.900	0.862
Panel B: Fall								
Post	-0.0170 (0.0888)	0.0763 (0.0579)	0.0708 (0.0535)	0.372* (0.178)	0.231 (0.143)	0.142 (0.0954)	0.132 (0.0874)	0.369 (0.307)
<i>N</i>	221	221	221	221	221	221	221	221
<i>R</i> ²	0.821	0.893	0.854	0.891	0.830	0.895	0.856	0.892
Quadratic	N	N	N	N	Y	Y	Y	Y

Notes: Regression coefficients using Equation (5). Dependent variable is log of metro ridership at each daytime period. Post refers to the period after the DST transition takes place in fall and spring. Sample size considers 3 weeks before and after DST transition. In panel A, we exclude days associated with Chilean national day. In panel B, we exclude February days since they correspond to summer vacations in Chile. Robust standard errors in parentheses.

* $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$.

We focus our attention on a single large urban area which allows us to extend the evidence of the effect of ambient light on crime in two important dimensions. Regarding the interpretation of the reduced-form coefficients, we highlight that our main estimates can be interpreted as responses associated with the interaction between supply and demand for offenses. Supply-side responses are related to potential offenders' actions and, in particular, to the extent that more (fewer) hours of ambient light during a certain time of day deter (stimulate) them to offend. On the other hand, a demand-side response refers to victims' behavior, which can subsequently alter the set of criminal opportunities available precisely because of the variation in ambient light. In theory, we expect that a variation in ambient light induced by DST policy may potentially affect both agents. Without specific information on agent behavior, we cannot be certain of the actual mechanism that is driving our results. For example, an extension of sunlight during 'sunset hours' can encourage people to spend more time outdoors, which may also potentially alter the protective measures potential victims adopt, which in turn can alter the behavior of offenders. We study a possible adaptive response associated with the shock in ambient light, specifically if the DST policy causes a substantial change in commuting patterns. We use Metro ridership data at the same frequency level of our previous estimates for the same period of analysis and we find no significant variation at any hour of day. Although subway ridership is a broad measure of victims' behavior, this finding suggests that the DST effect is unlikely to be driven by an endogenous reaction in commuting patterns, which is arguably a key indicator of victim behavior.

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Supplementary material

[Supplementary data](#) for this paper are available at *Journal of Economic Geography* online.

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